ECE 447

Lesson 19
Pulse Code Modulation



UNITED STATES
AIR FORCE
ACADEMY

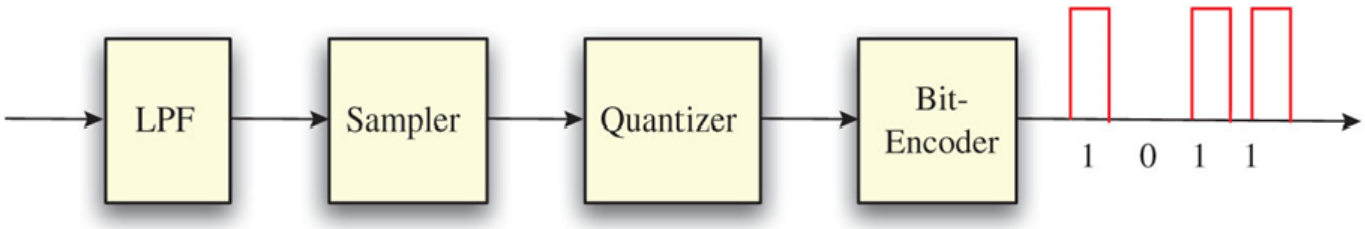
Schedule and Admin

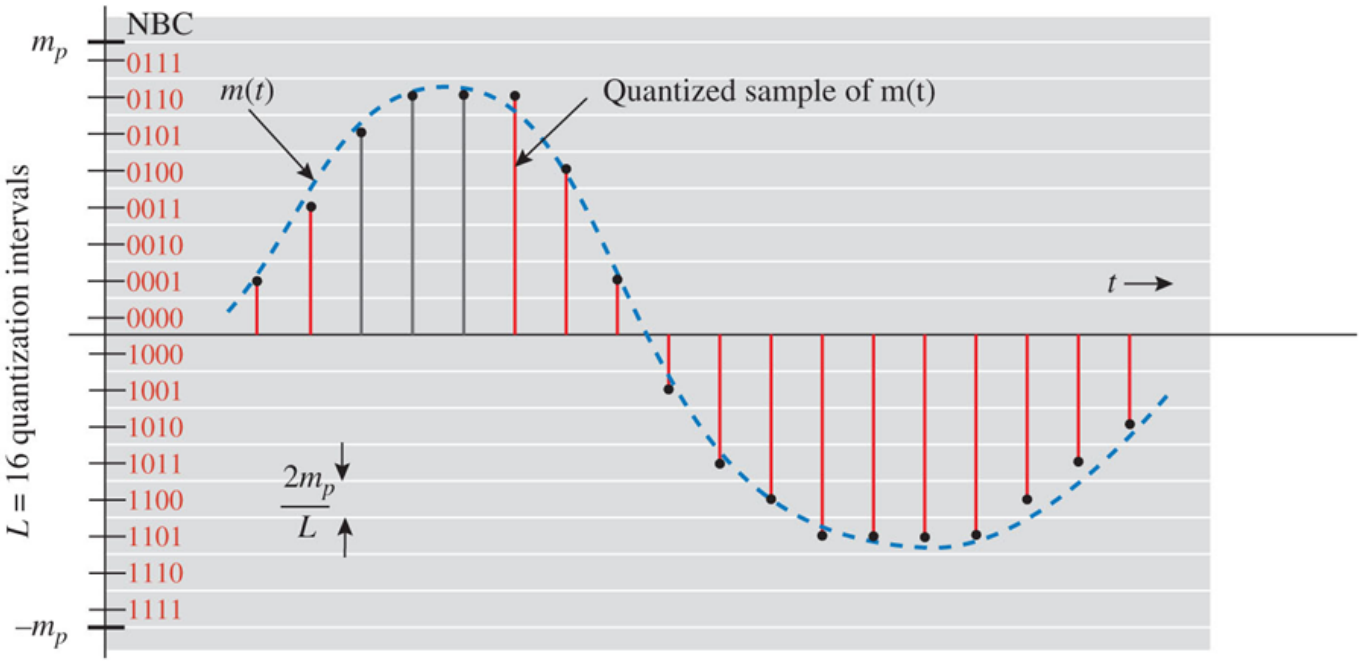
- Schedule
- Admin
  - **Lab 4**. Due tonight (Lesson 19, 30 Sep) to Gradescope.
  - **HW4**. Due Lesson 21 (Tue, 6 Oct) to Gradescope.
  - **Lab 5**. Begins Lesson 21 (Tue, 6 Oct); due Lesson 24 (Thu, 15 Oct).

Example Problem - 5.1-7

Sampling Computer Example

Pulse Code Modulation





Quantization Error and SNR

$$\begin{aligned}\left(\frac{S_o}{N_o}\right)_{\text{dB}} &= 10\log_{10}\left(\frac{S_o}{N_o}\right)\\ &= 10\log_{10}\left[c(2)^{2n}\right]\\ &= 10\log_{10} c + 2n\log_{10} 2\\ &= (\alpha + 6n) \quad \text{dB}\end{aligned}$$

PCM Computer Example